

Surya Vardhan

Senior Engineer

Firmware engineer with 5+ years of experience delivering four commercial products from concept to market with a 35% reduction in development time. Expert in developing RTOS-based drivers using C, with strong command over protocols (I2C, SPI, CAN) and embedded Linux. Proactive, collaborative, and passionate about system-level optimization.

WORK EXPERIENCE

Senior Engineer 12/2023 - Present

Bosch Global Software Technologies Pvt Ltd

- Developed Python applications for execution in an Embedded Linux environment.
- Performed unit testing using the pytest framework.
- Created a battery monitoring application in Python for State of Charge (SOC) and State of Health (SOH) calculations.
- Developed DBus applications for Inter-Process Communication (IPC) between multiple applications on Linux-based devices.
- Design and development of new-gen Bosch Home appliance electronics [Vacuum Cleaners, Robotic Cleaners] including EOL.

Senior Software Design Engineer 04/2022 - 12/2023

Mistral Solutions Pvt Ltd

- Involved in full SDLC processes including requirements, design, coding, unit testing, Static analysis, releases, debugging, maintenance, and documentation.
- Followed agile methodology and actively taken part in scrum calls and retrospection meetings.
- Ported MMCSO device driver for AM263x Sitara-based microprocessors using TI Code Composer Studio in 3 months. Awarded Pride Pack for meeting SDK delivery timelines for AM68/AM69
- Accelerated test farm integration by 25% for Jacinto processor-based development boards for automation testing in the Linux environment.
- Experienced in using JIRA for Bug Tracking, Bitbucket, Confluence, and Git for Version control.
- Created board-specific boot binaries for AM68 using device tree files for Linux-based board bring-up and interface integration.
- Contributed to technical recruitment efforts for hiring tech experts.

Embedded Software Engineer 07/2021 - 02/2022

Nash Tech Labs

- Led a team of 4+ peers in hardware and software integration for the development of on-board chargers for electric vehicles.
- Traced circuits, analyzed system requirements, and debugged software for test logs and summary reports.
- Developed 32-bit MCU firmware using IAR and STM32 Cube IDE for 3.3kW electric vehicle charging station.
- Created technical documents including software requirements, software

CONTACT

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SKILLS

Programming Language / Scripts:

- Embedded C (Advanced)
- C++ (Intermediate)
- Shell & Python
- Assembly Code debugging
- Bare Metal Coding

Environment Tools:

- STM32 Cube IDE
- IAR IDE
- Keil IDE
- ESP-IDF
- Code Composer Studio
- VSCode
- Make File

Operating Systems:

- RTOS [FreeRTOS & TI-RTOS]
- Embedded Linux
- Multithreading & IPC
- SoC Boot Sequence

Communication Protocols:

- I2C
- SPI
- UART, USART
- CAN
- BLDC Motor Control

Hardware Debugging:

- Oscilloscope
- Multimeter
- Logic Analyzer
- JTAG, SWD
- CANoe / CANalyzer
- Spectrum Analyzer
- Schematic Design
- Soldering LQFP Pkg IC

MCU & MPU:

- 8051, ATmega328
- STM32 [ARM Cortex M0+, M3 & M4]
- ESP32 & ESP8266
- AM263, AM68 & AM69 [A57 & M4]
- Teensy [NXP - MK64, MK66, iMXRT1062]

Version Control & Tools:

- GitHub
- Jira
- Bit Bucket
- Confluence

design descriptions, software verification, datasheets, PCB verification, user manuals, whitepapers, and brochures.

- Managed a team of four peers in the design and development of consumer electronics products.
- Recognized as the fastest team leader in the organization for completing a RTOS-based project from concept to commercialization within 4 months.

Senior Embedded Engineer

08/2019 - 06/2021

Senpronics

- Developed RTOS-based C code in compliance with Misra standards for security devices and products.
- Achieved 10% higher and stable performance in IOT-based product development using multithreading and multiprocessing techniques, resulting in a 30% increase in orders.
- Reduced development costs by 25% by designing Windows software for communicating with MCUs over serial communication protocol using LabVIEW modeling tool.
- Materialized mission-critical products like bomb detonator jammers for the Indian Army using FreeRTOS with IPC and multithreading concepts, increasing revenue by 30%.
- Pioneered high-bandwidth Wi-Fi transmission for high-precision DAQ.
- Improved firmware with a 20% increase in response time while gaining knowledge of radio frequency technology in the development of RF signal generators.
- Proposed proper technical solutions in Scrum meetings, based on in-depth knowledge of 16-bit and 32-bit microcontroller/microprocessor ARM architecture, resulting in a 30% increase in project velocity.

CERTIFICATION

Advanced Embedded Systems

07/2019

Vector India Org - Chennai

EDUCATION

Bachelor of Engineering

05/2017

Anna University - Chennai [Affl]

Coimbatore Institute of Engineering and Technology

Field of Study in Mechatronics

INTERNSHIP

Defense Research and Development Organization [DRDO]

11/2016

Combat Vehicle Research and Development [CVRDE] - Chennai

Final Year Internship for 6 Months.

Involved in prototype development of Regenerative Braking System for ARJUN Mark-2 Main Battle Tank [MBT].

PATENT

Automated Security System and Method for Motor Bike Locking

09/2016

Application number:20164103054

Issue: 09/2016

International Classification: G08B21/00

ACHIEVEMENTS

Pride Pack (Outstanding Performance)

For: Excellent efforts and teamwork in making AM68 & AM69 SK SDK release meeting customer schedule.

By: Mistral Solutions Pvt Ltd.

LANGUAGES

- English (Fluent)
- Hindi (Fluent)
- Tamil (Native)
- Kannada (Conversational)
- Telugu (Conversational)
- German (A1)

SOFT SKILLS

- Effective communication
- Good Collaborator
- Initiative-taker
- Self-driven.
- Tech Inspired
- Result-oriented
- Continuous learner
- People skills
- Adaptability
- Cheerful outlook
- Meticulous

PROJECTS

- [TI AM64x Inter-core IPC for ADC data transfer \(R5 ↔ A53\)](#)
- [Naïve Bayes Estimator on ARM Cortex M0 \(Bare-Metal C\)](#)
- [TCP Graph plotter \(Matplotlib\)](#)
- [Hi-Speed data transfer over TCP on ESP32](#)
- [ESP32 Based Velocity Tracker](#)
- [RF Synthesizer](#)
- [HMI With STM32](#)
- [CAN N/W on STM32](#)
- [Regenerative Braking system for Battle tank using BLDC Motors](#)

PRODUCTS

- [Earthquake Detector \[EMD - 1202®\]](#)
- [Portable Seismic DAQ \[PS-DAQ®\]](#)
- [Real Time Acceleration Plotter \[RTAP-335®\]](#)
- [Remote Tree Climber](#)
- [Devantech ETH008-B Driver](#)